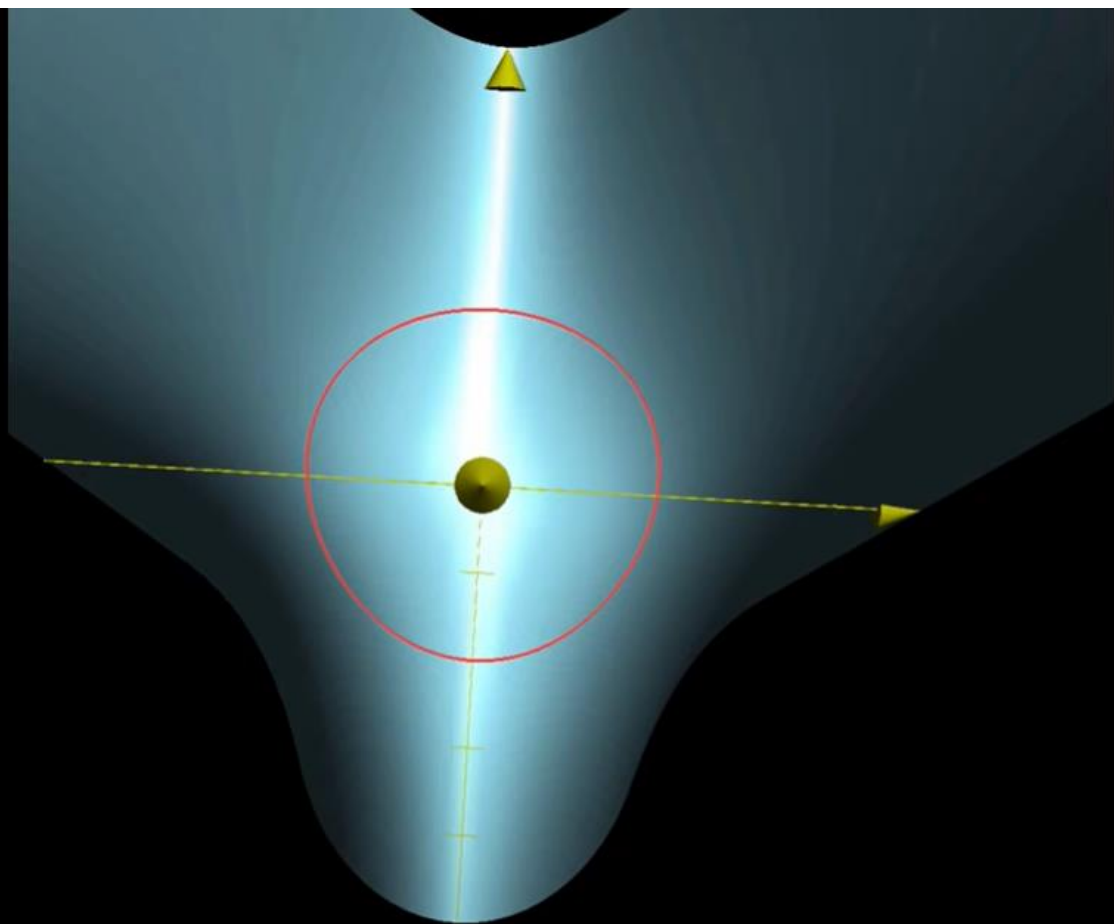


Constrained Optimization

Maximize $f(x,y) = x^2y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$





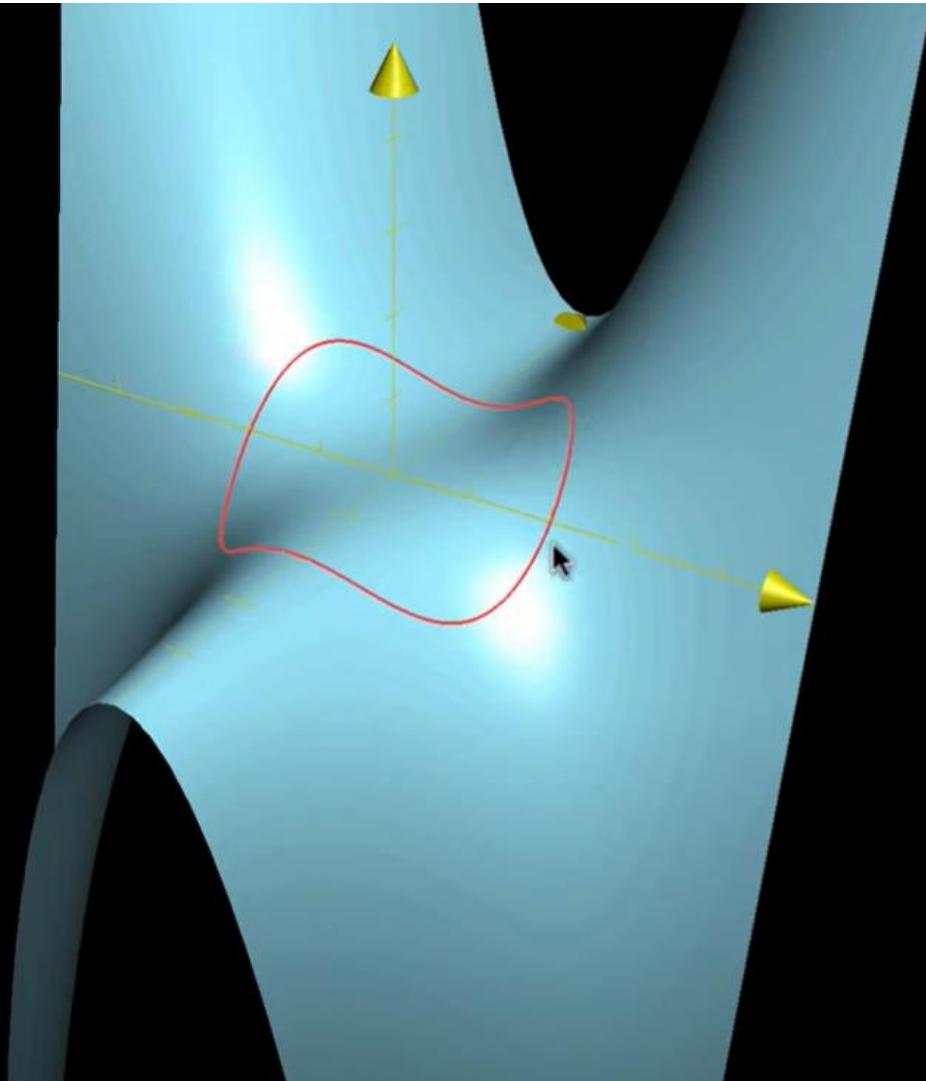
Constrained Optimization

Maximize $f(x, y) = x^2y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$

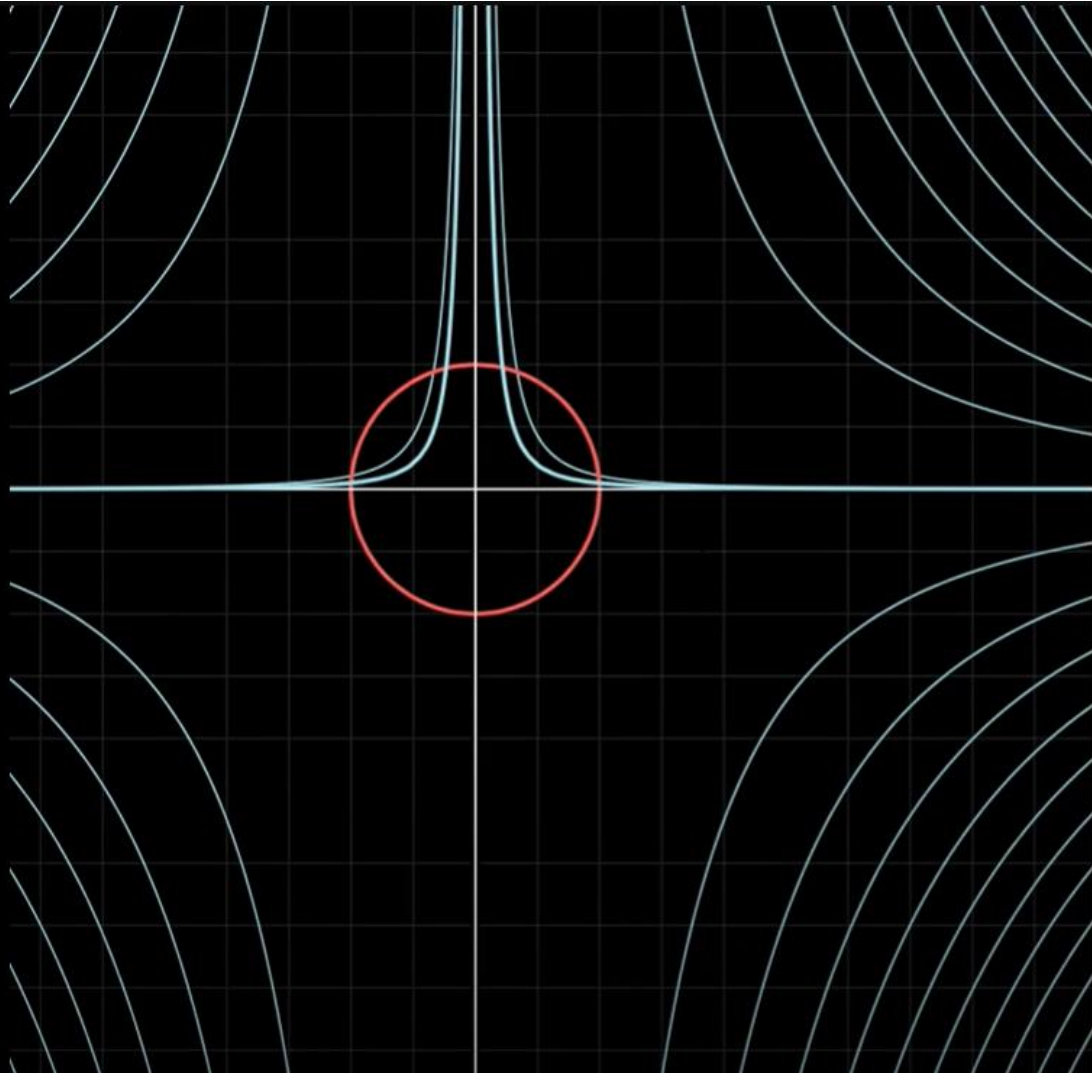
Constrained Optimization

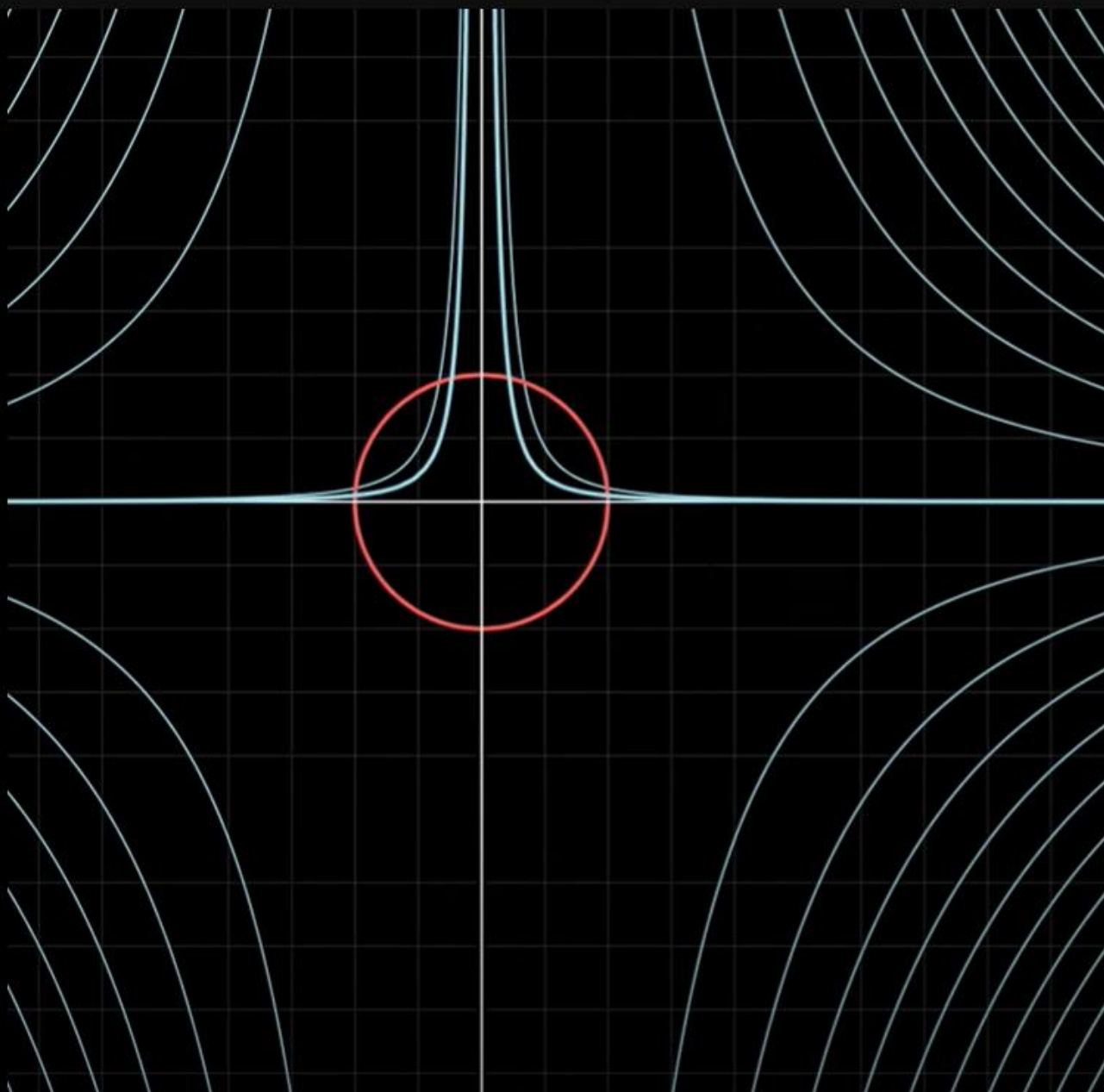
Maximize $f(x,y) = x^2y$
on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$



Constrained Optimization

Maximize $f(x,y) = x^2y$
on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$



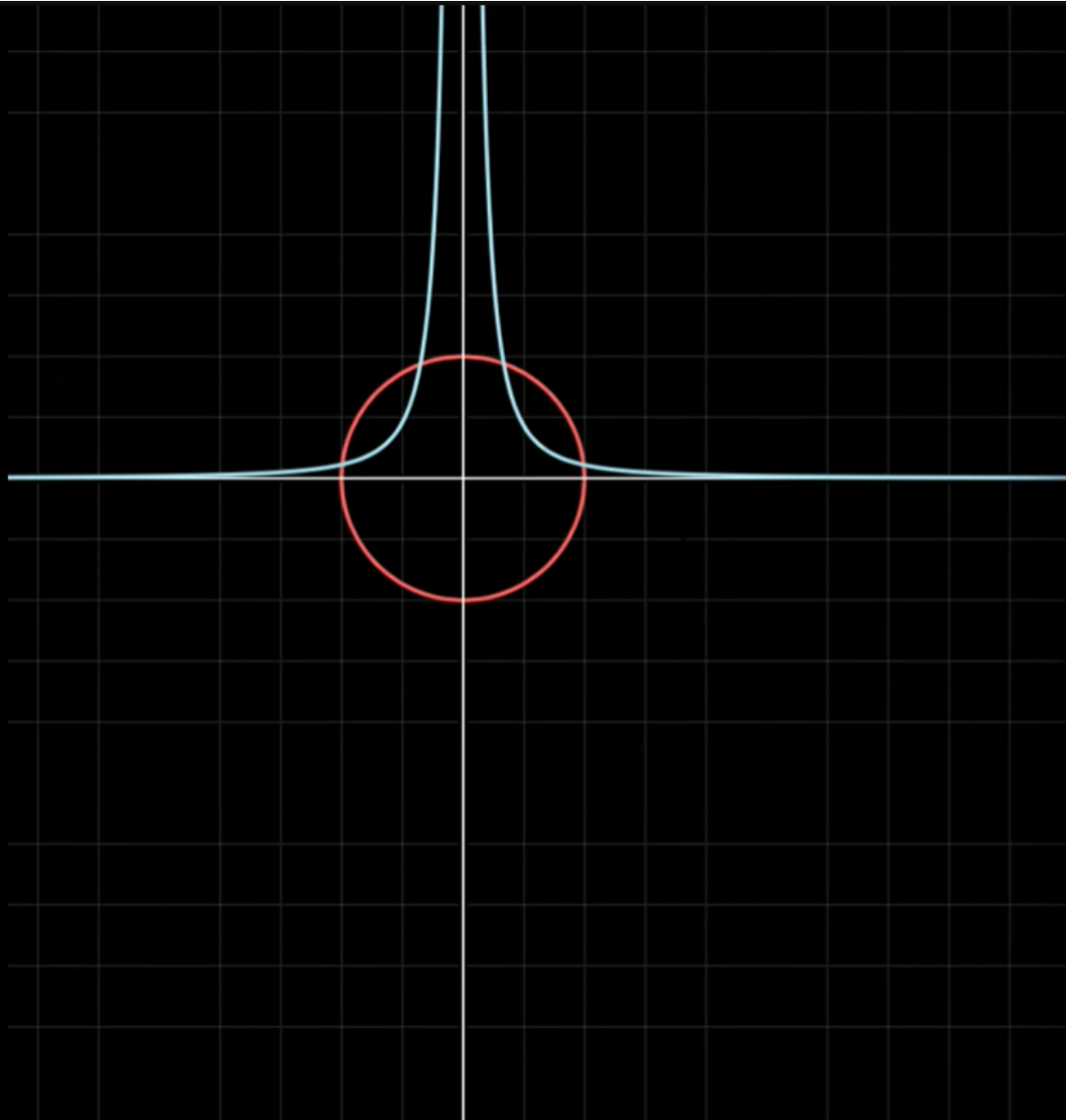


Constrained Optimization

Maximize $f(x, y) = x^2 y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$

$$f(x, y) = \underline{\underline{z}}$$



Constrained Optimization

Maximize $f(x, y) = x^2 y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$

$$f(x, y) = 0.1$$

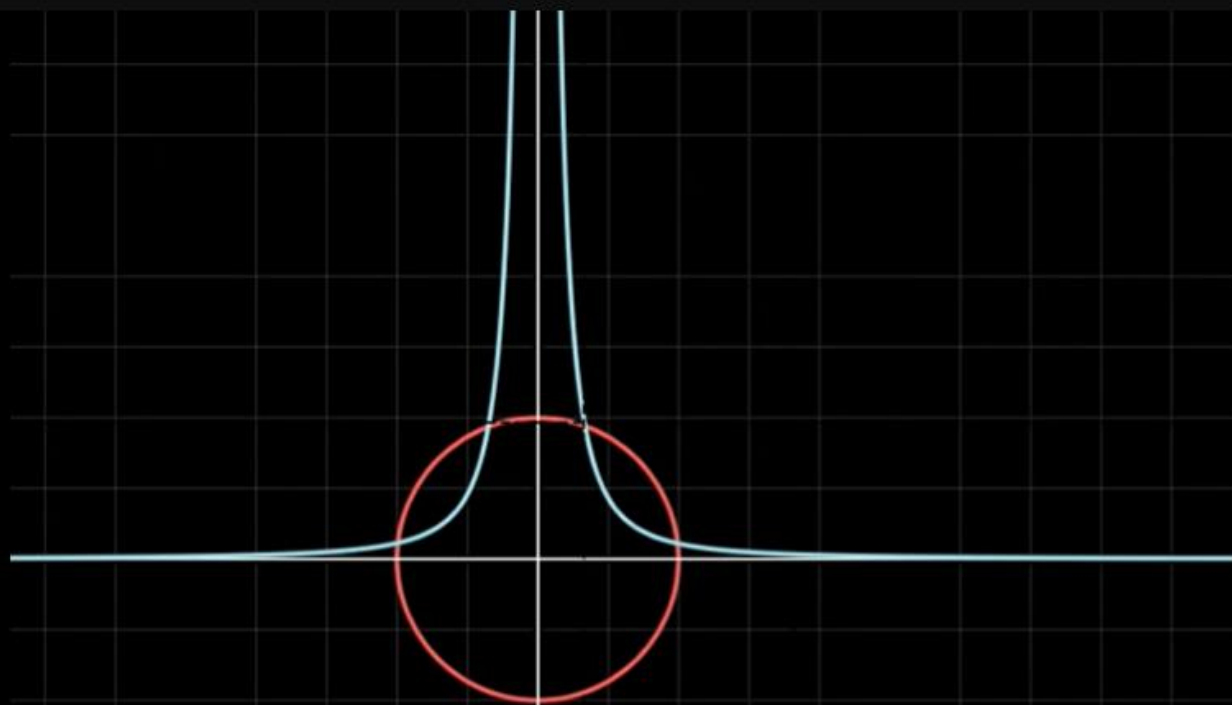


Constrained Optimization

Maximize $f(x, y) = x^2 y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$

$$f(x, y) = 0.1 \quad | \quad f(x, y) = 1$$



Constrained Optimization

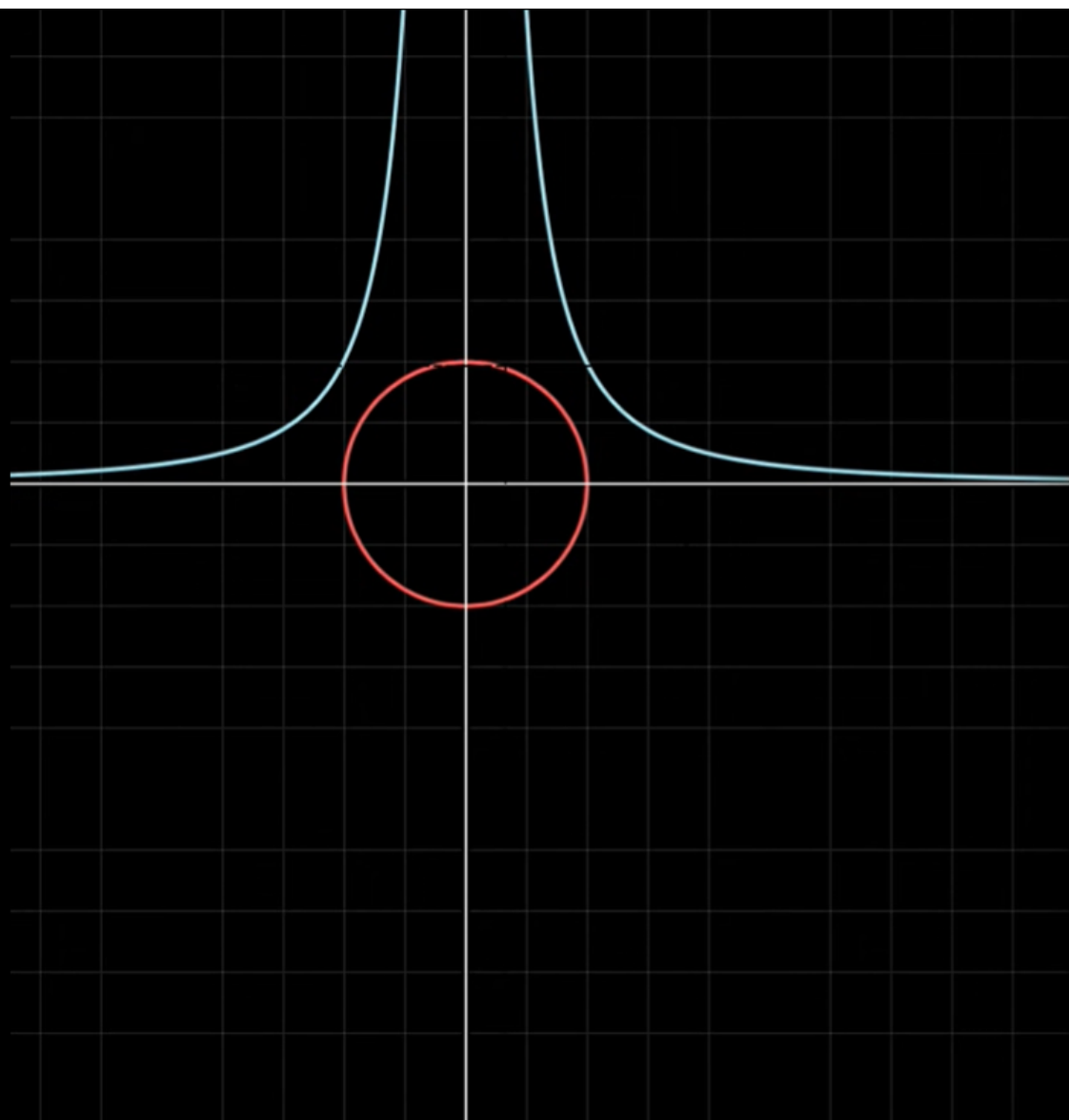
Maximize $f(x, y) = x^2 y$
on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$

$$f(x, y) = 0.1$$

$$(x, y) \nearrow \checkmark$$

$$x^2 + y^2 = 1$$

$$f(x, y) = 1$$



Constrained Optimization

Maximize $f(x,y) = x^2y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{unit circle}}$

$$f(x,y) = 0.1$$

$(x,y) \nearrow \checkmark$

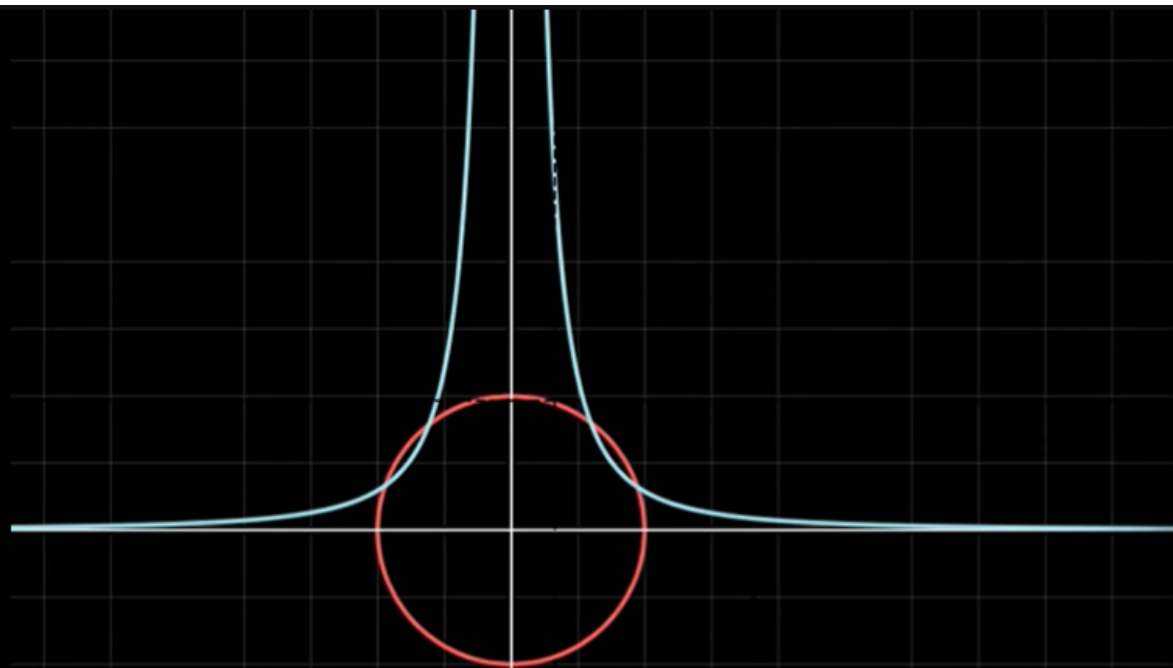
$$x^2 + y^2 = 1$$

$$f(x,y) = 1$$

$(x,y) \nearrow$

$$\text{off } x^2 + y^2 = 1$$





Constrained Optimization

Maximize $f(x, y) = x^2 y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{Unit circle}}$

$$f(x, y) = \overset{+}{0.3} \cancel{0.1}$$

$(x, y) \nearrow \checkmark$

$$x^2 + y^2 = 1$$

$$f(x, y) = 1$$

$(x, y) \nearrow$

$$\text{off } x^2 + y^2 = 1$$



Constrained Optimization

Maximize $f(x, y) = x^2 y$

on the set $\underbrace{x^2 + y^2 = 1}_{\text{Unit circle}}$

$$f(x, y) = \overset{0.3}{\cancel{0.1}}$$

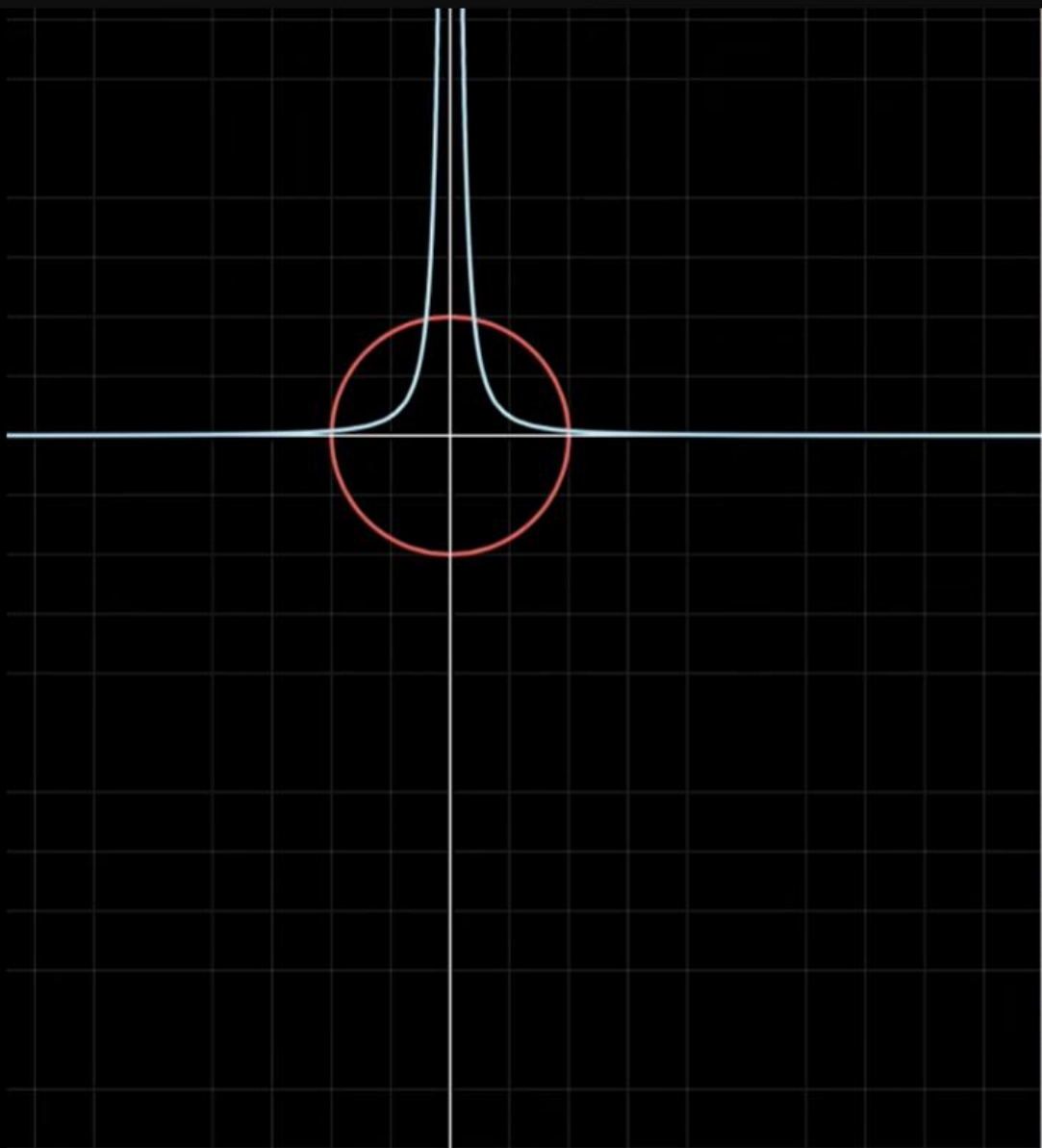
$(x, y) \nearrow \checkmark$

$$x^2 + y^2 = 1$$

$$f(x, y) = 1$$

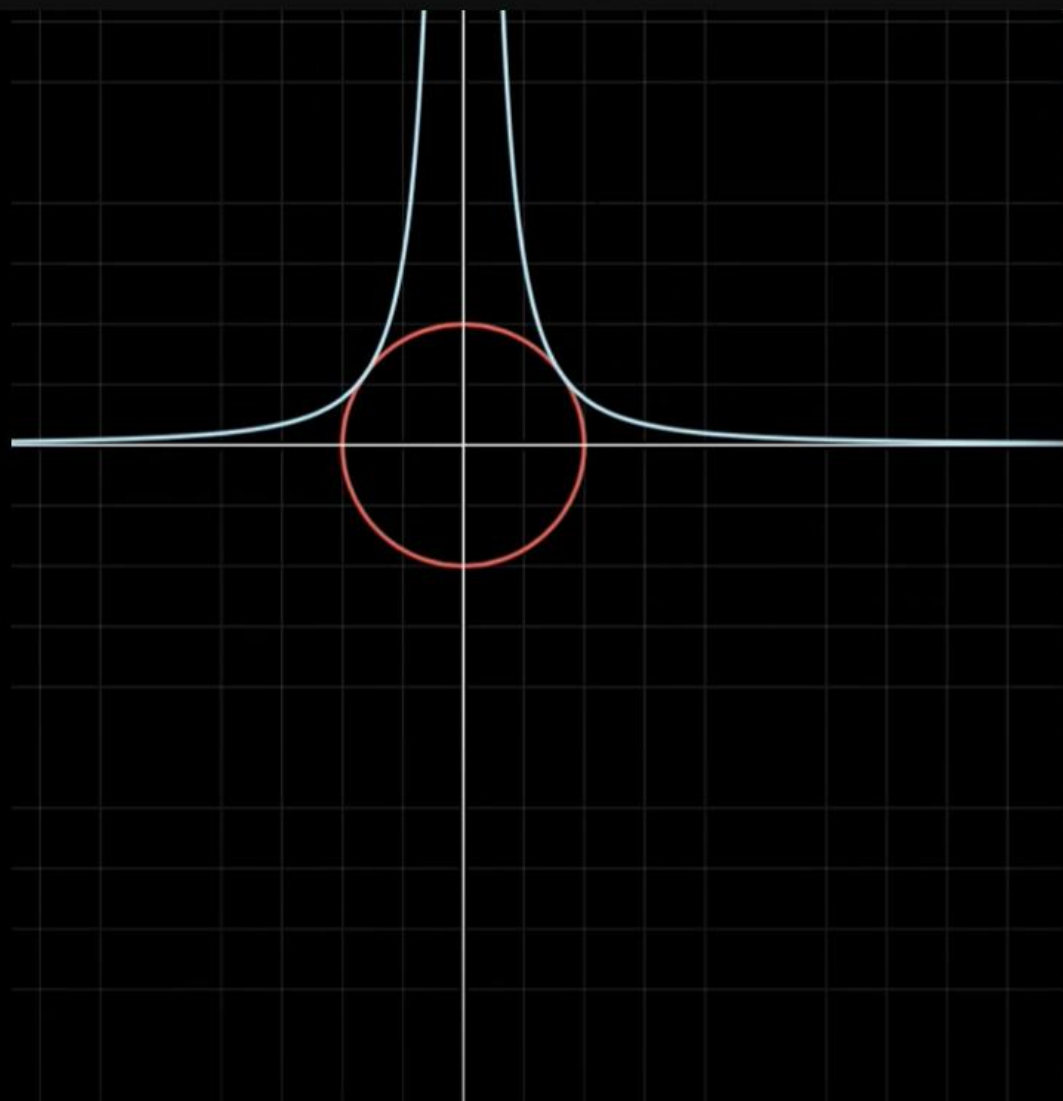
$(x, y) \nearrow$

$$\text{off } x^2 + y^2 = 1$$



Maximize $f(x, y) = x^2 y$
on the set $x^2 + y^2 = 1$

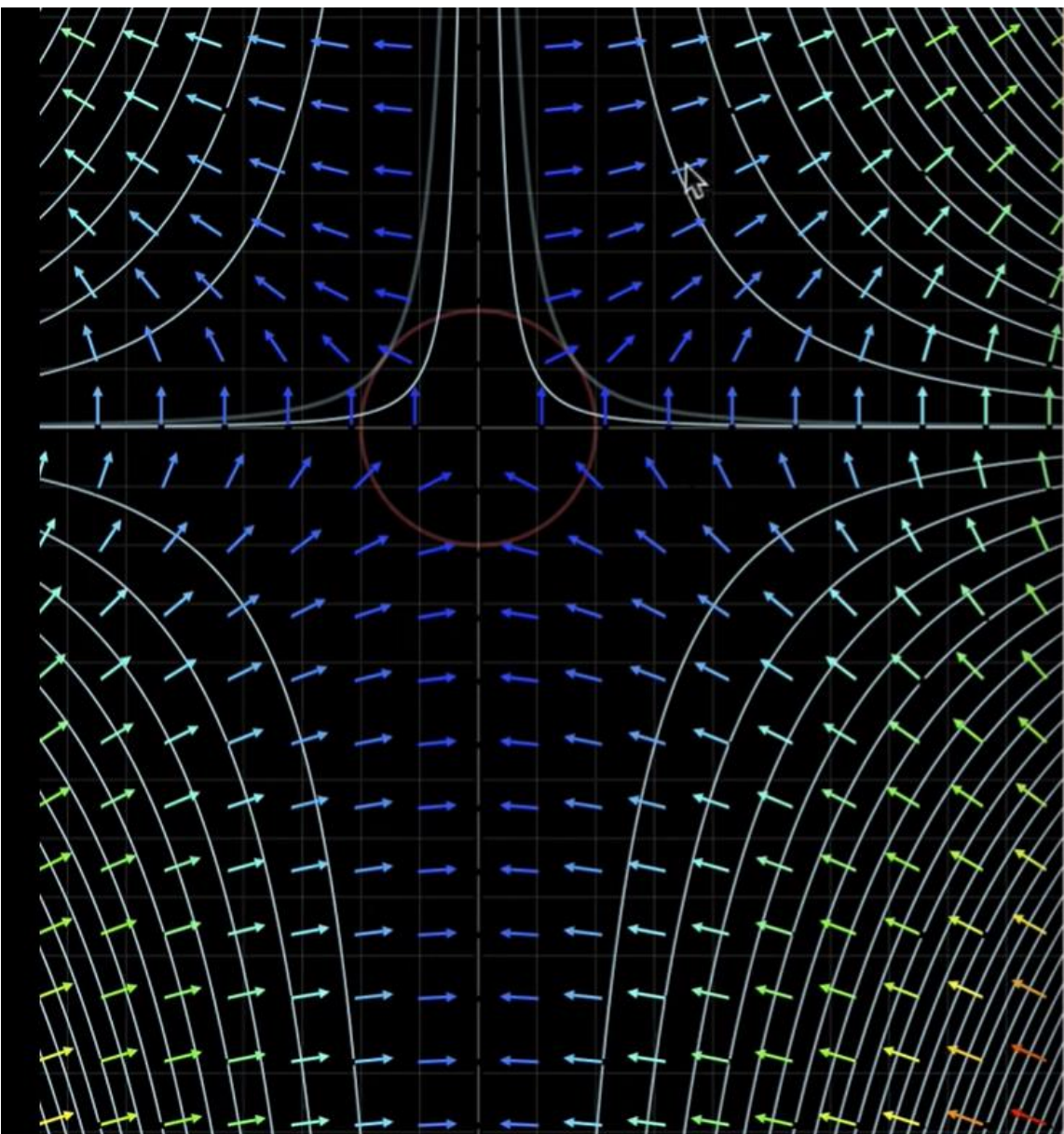
$$f(x, y) = c$$



Maximize $f(x, y) = x^2y$
on the set $x^2 + y^2 = 1$

$$f(x, y) = c$$

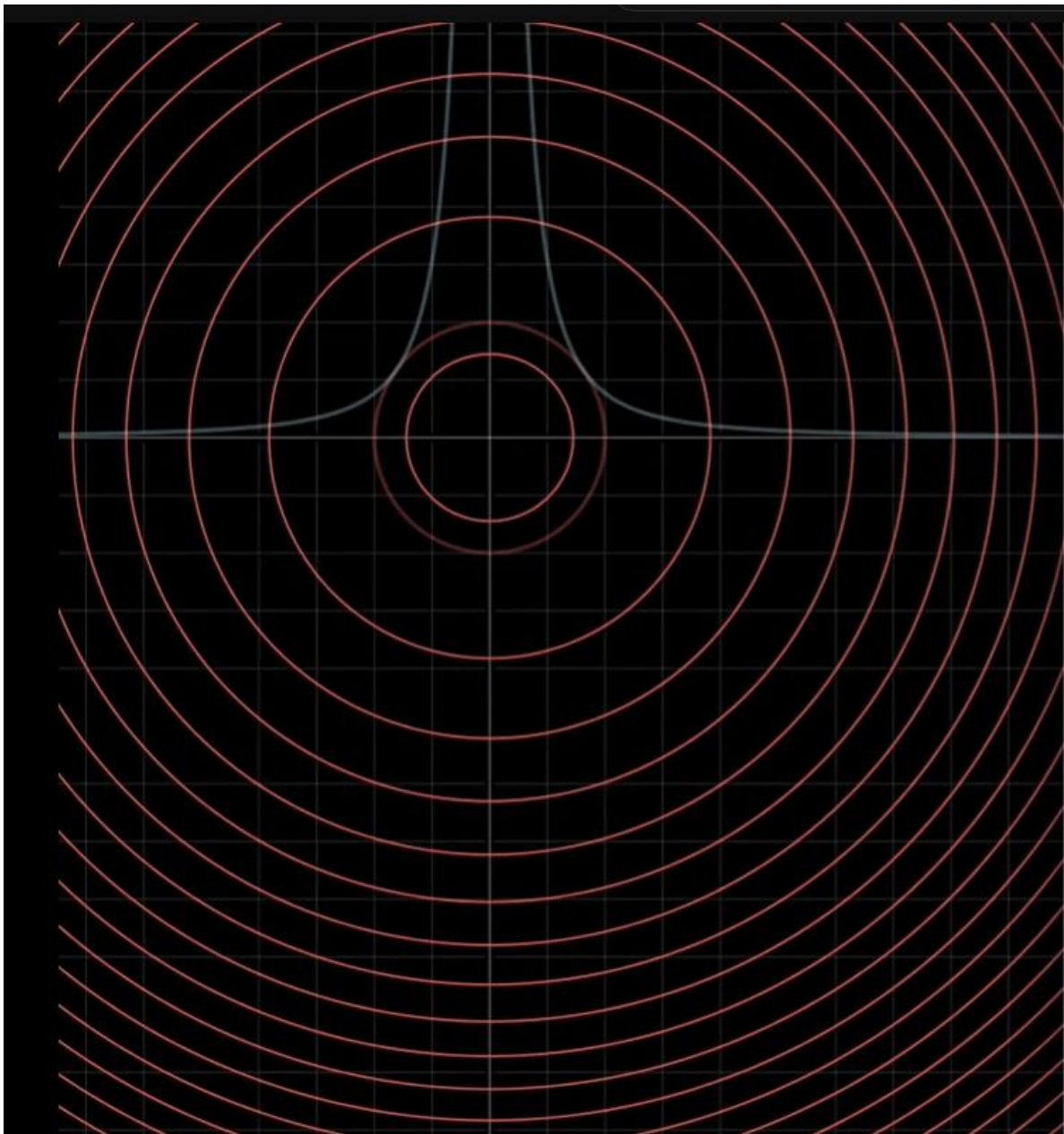




Maximize $f(x,y) = x^2y$
on the set $x^2 + y^2 = 1$

$$f(x,y) = c$$
$$\nabla f$$



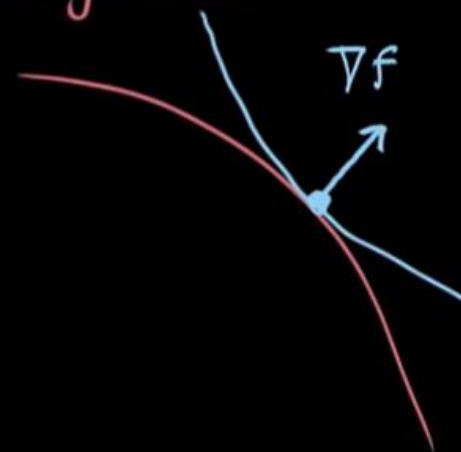


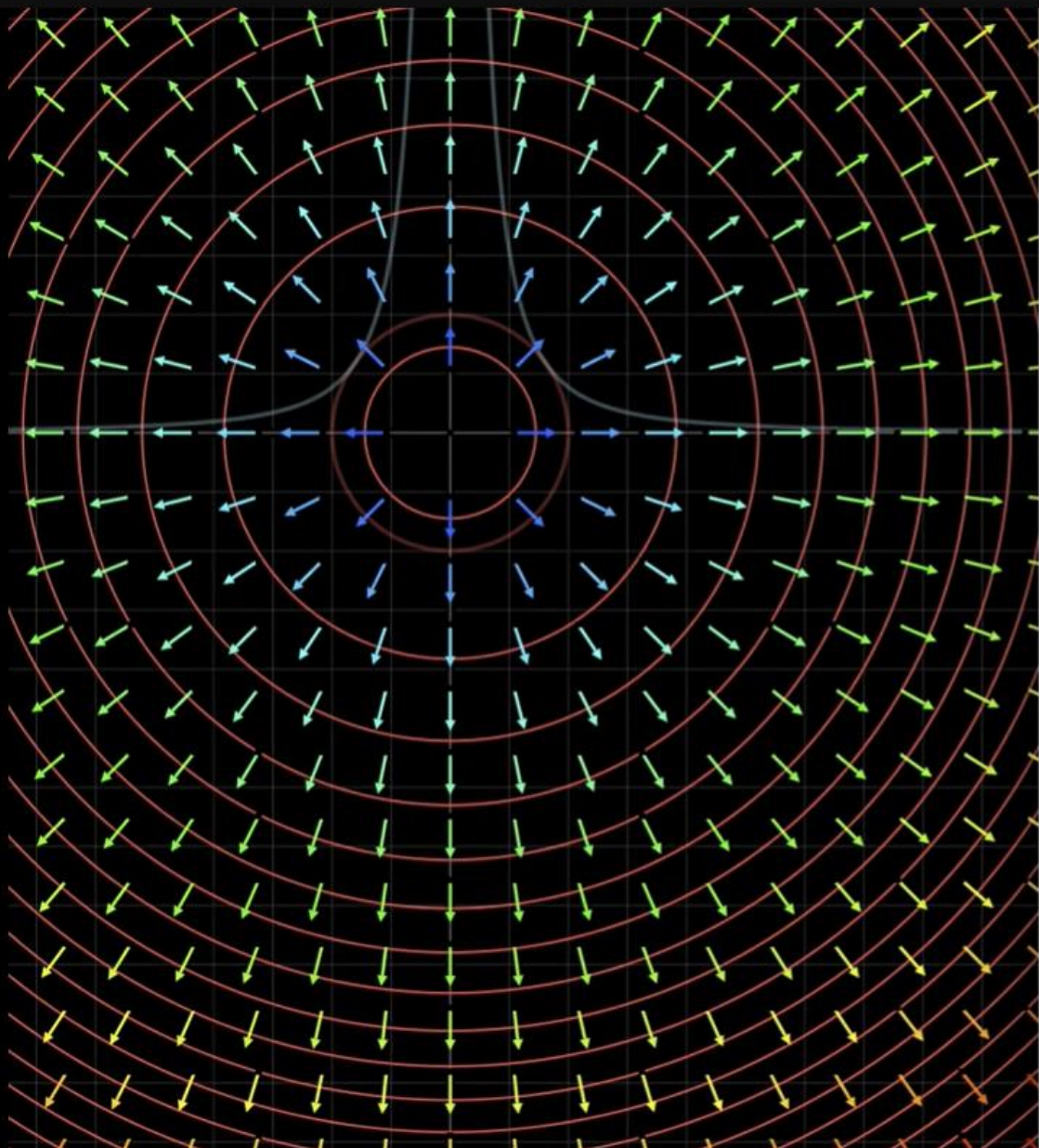
Maximize $f(x, y) = x^2 y$
on the set $x^2 + y^2 = 1$

$$f(x, y) = c$$

$$\nabla f$$

$$g(x, y) = x^2 + y^2$$





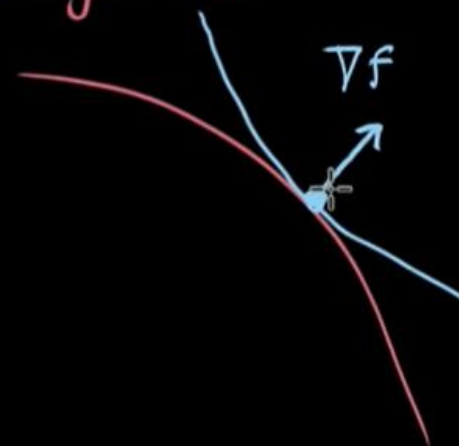
Maximize $\underline{f(x, y) = x^2 y}$
on the set $\underline{x^2 + y^2 = 1}$

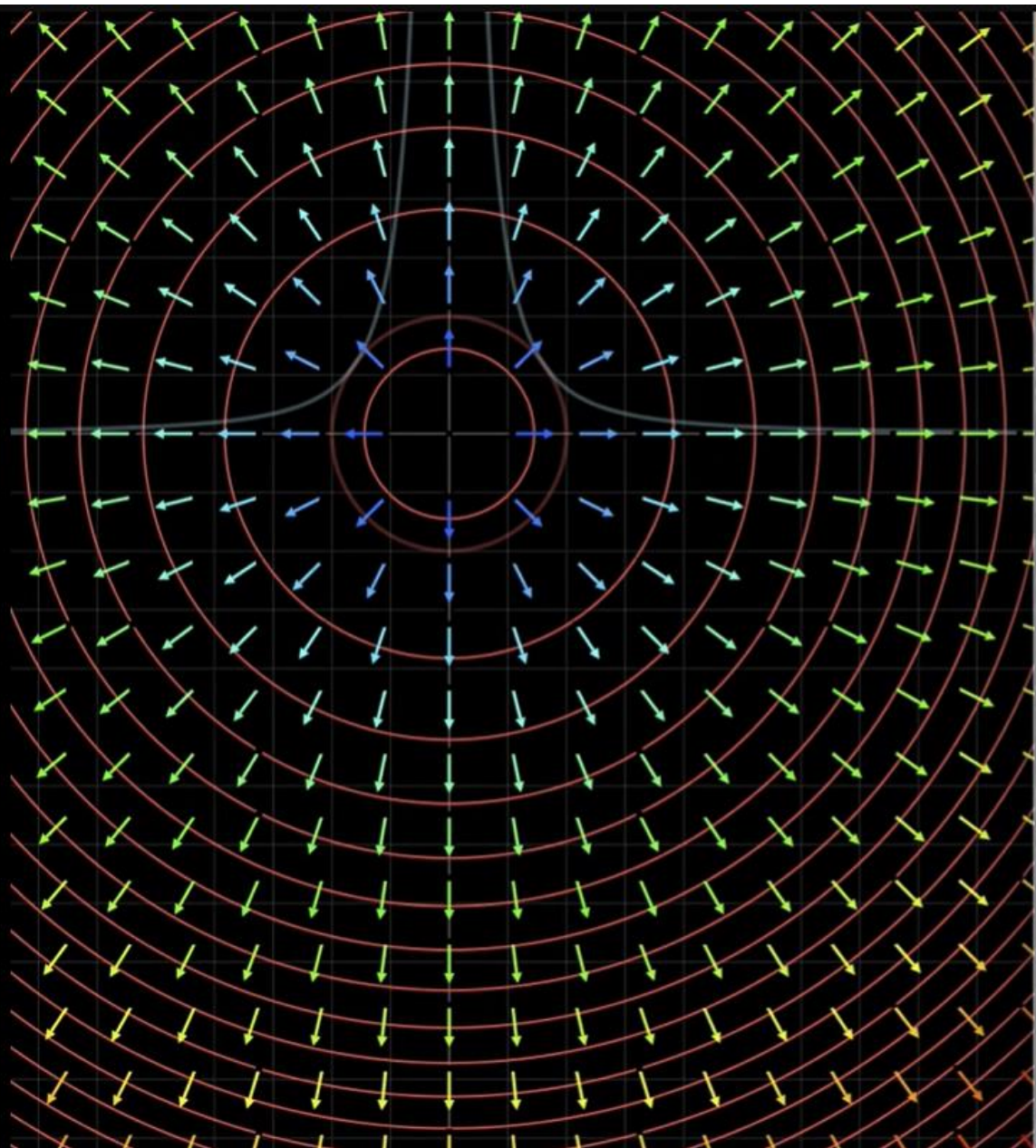
$$f(x, y) = c$$

$$\nabla f$$

$$g(x, y) = x^2 + y^2$$

$$\nabla g$$





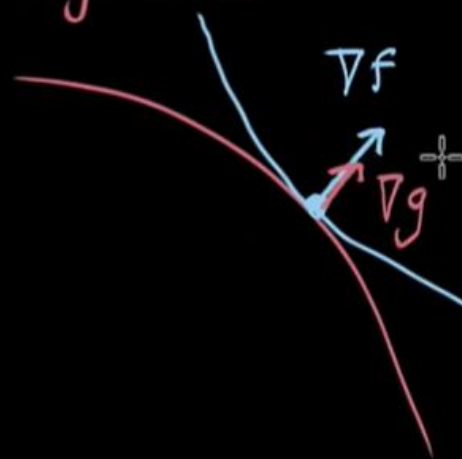
Maximize $f(x, y) = x^2 y$
on the set $x^2 + y^2 = 1$

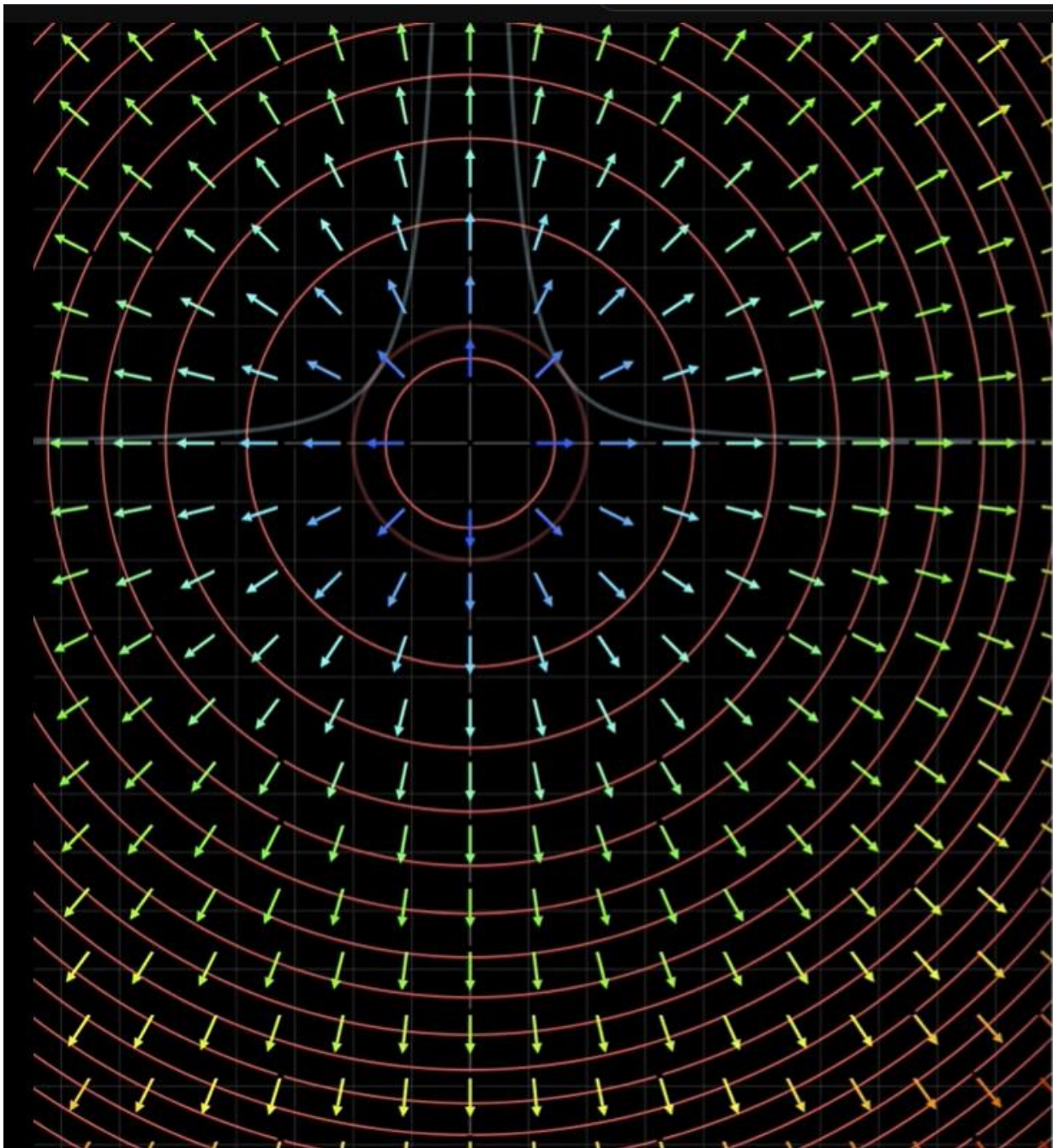
$$f(x, y) = c$$

$$\nabla f$$

$$g(x, y) = x^2 + y^2$$

$$\nabla g$$





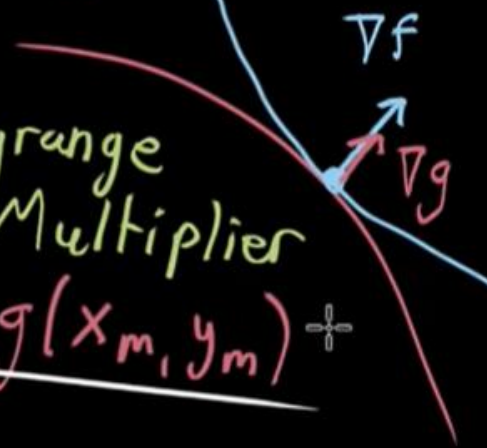
Maximize $f(x, y) = x^2 y$
on the set $x^2 + y^2 = 1$

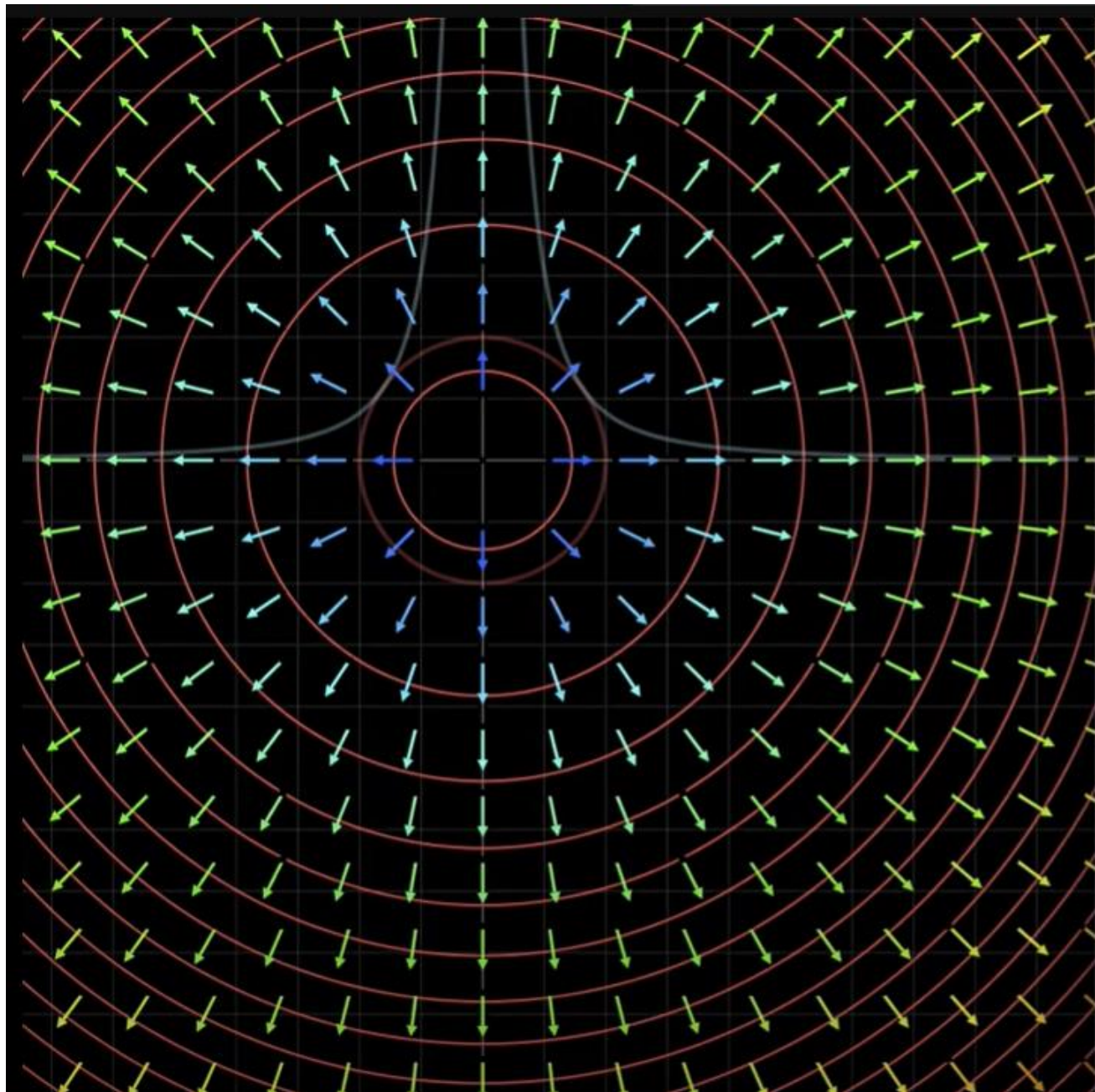
$f(x, y) = c$ Lagrange
Multiplier

$$\nabla f(x_m, y_m) = \lambda \nabla g(x_m, y_m) +$$

$$g(x, y) = x^2 + y^2$$

∇g



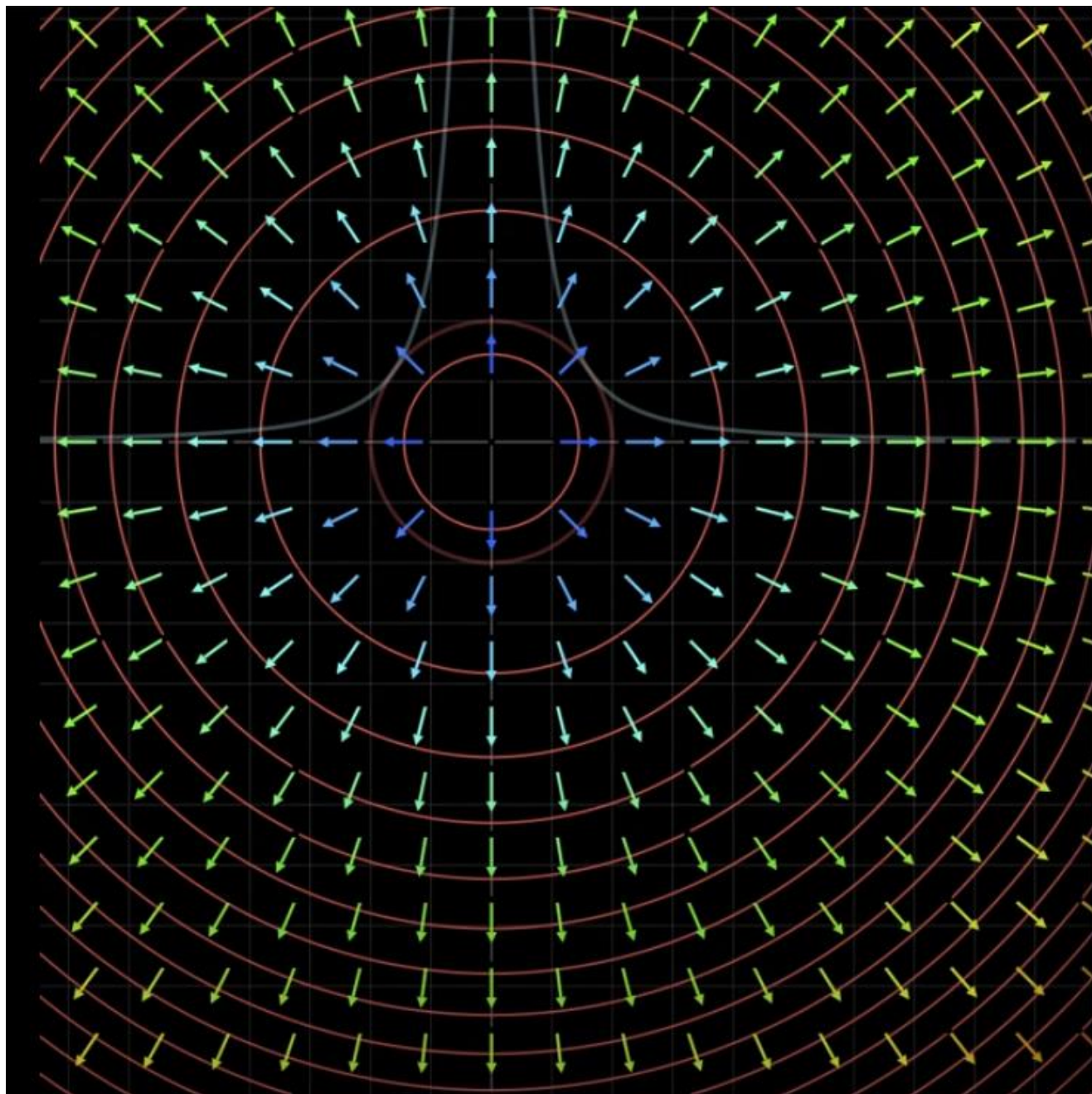


$$\nabla f(x_m, y_m) = \lambda \nabla g(x_m, y_m)$$

Multiplier

$$g(x, y) = x^2 + y^2$$

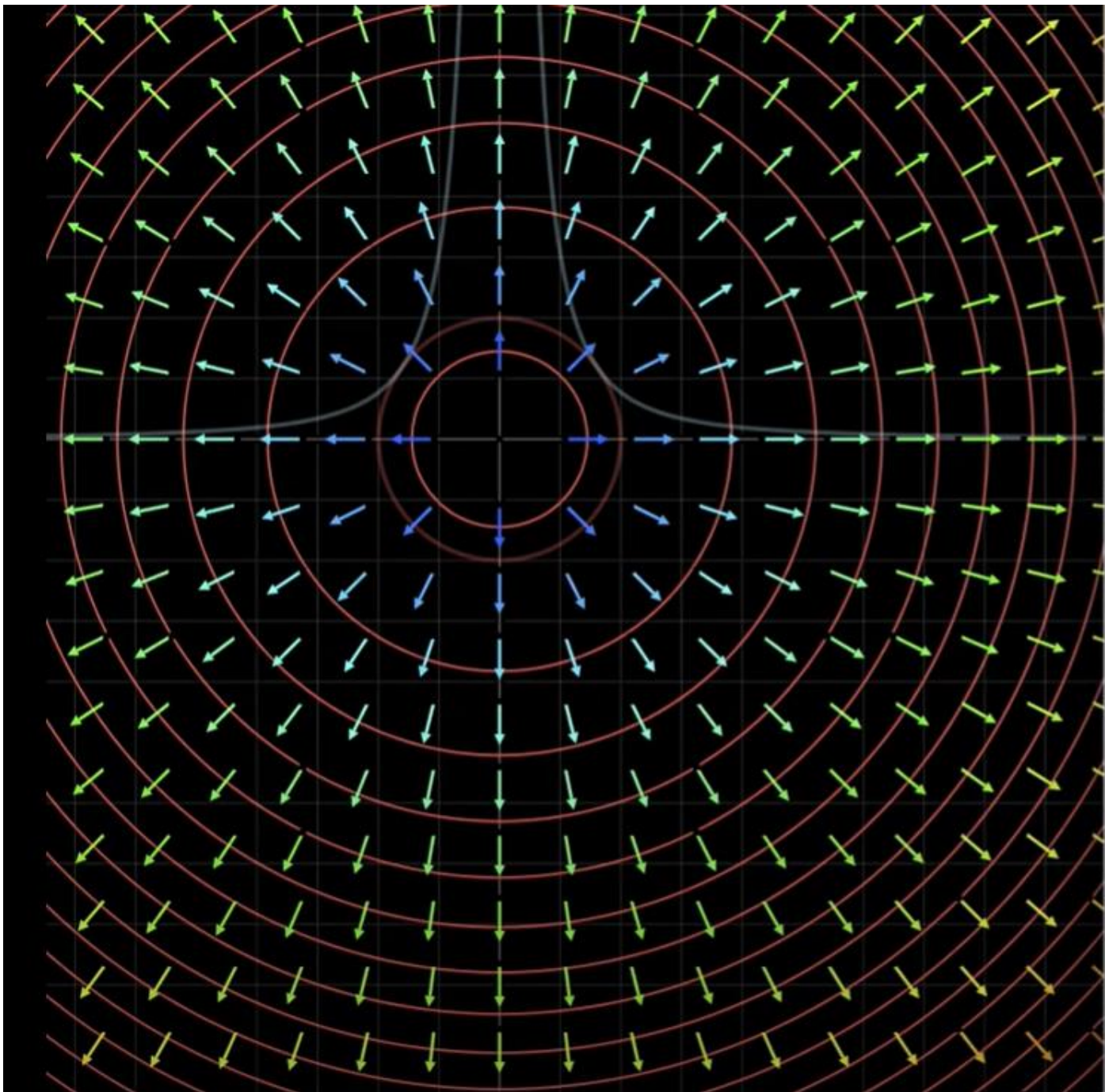
$$\nabla g = \nabla(x^2 + y^2) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$



$$\nabla g = \nabla(x^2 + y^2) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$\nabla(x^2 y) = \begin{bmatrix} 2xy \\ x^2 \end{bmatrix}$$

$$\begin{bmatrix} 2xy \\ x^2 \end{bmatrix} = \lambda \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

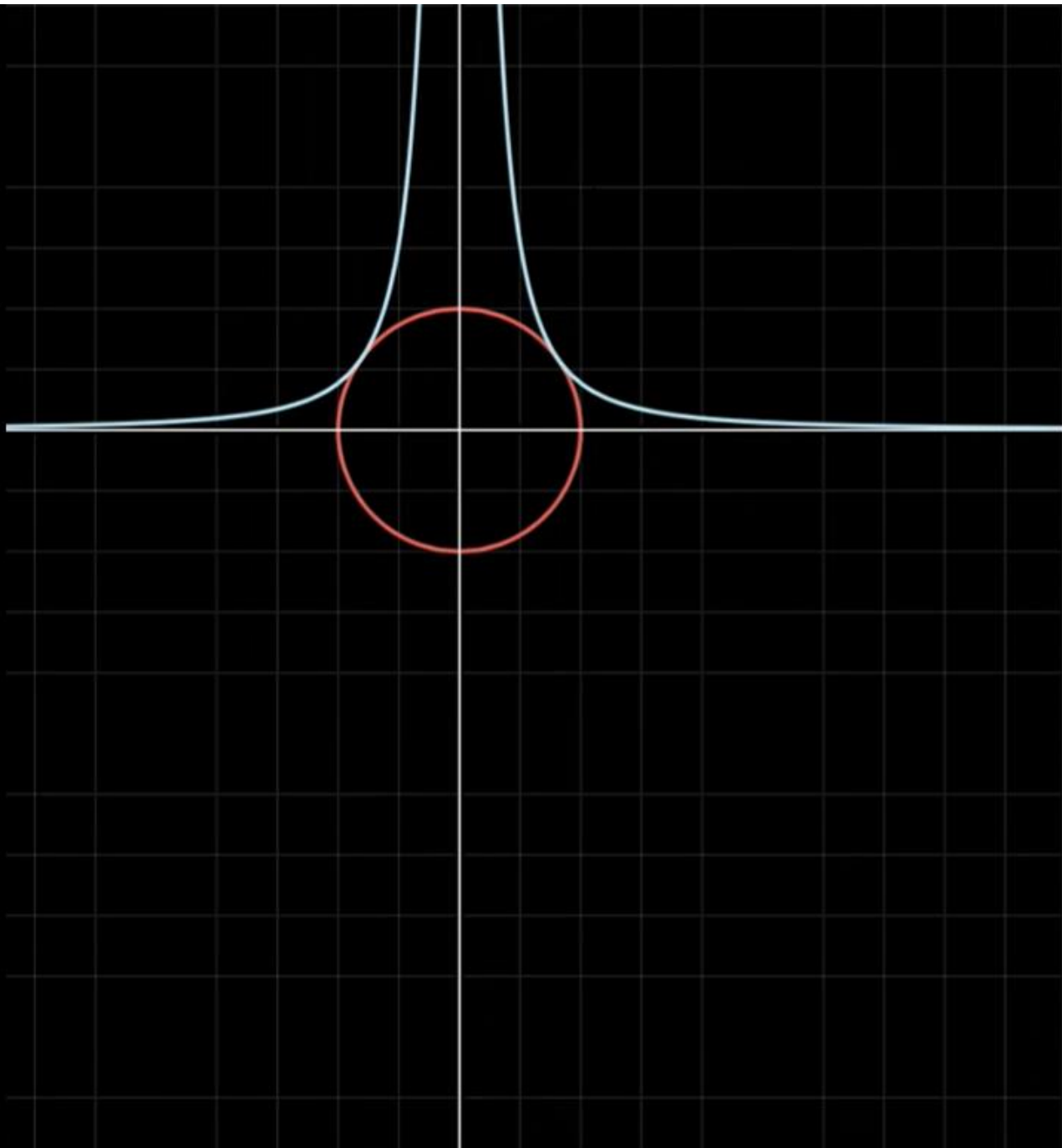


$$\begin{bmatrix} 2xy \\ x^2 \end{bmatrix} = \lambda \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$2xy = \lambda 2x$$

$$x^2 = \lambda 2y$$





$$\begin{bmatrix} 2xy \\ x^2 \end{bmatrix} = \lambda \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$2xy = \lambda 2x$$

$$x^2 = \lambda 2y$$

$$x^2 + y^2 = 1$$

$$x \neq 0$$

$$2xy = \lambda 2x \rightarrow 2y = \lambda 2 \rightarrow y = \lambda$$

$$x^2 = \lambda 2y \rightarrow x^2 = 2y^2$$

$$x^2 + y^2 = 1 \rightarrow 2y^2 + y^2 = 1 \rightarrow 3y^2 = 1 \rightarrow y^2 = \frac{1}{3}$$

$$y = \pm \sqrt{\frac{1}{3}}$$

$x \neq 0$

$$2xy = \lambda 2x \rightarrow 2y = \lambda 2 \rightarrow y = \lambda$$

$$x^2 = \lambda 2y \rightarrow x^2 = 2y^2 = 2 \cdot \frac{1}{3} \rightarrow x = \pm \sqrt{\frac{2}{3}}$$

$$x^2 + y^2 = 1 \rightarrow 2y^2 + y^2 = 1 \rightarrow 3y^2 = 1 \rightarrow y^2 = \frac{1}{3}$$

$$y = \pm \sqrt{\frac{1}{3}}$$

(x ≠ 0)

$$2xy = \lambda 2x \rightarrow 2y = \lambda 2 \rightarrow y = \lambda$$

$$x^2 = \lambda 2y \rightarrow x^2 = 2y^2 = 2 \cdot \frac{1}{3} \rightarrow \boxed{x = \pm \sqrt{\frac{2}{3}}}$$

$$\underline{x^2 + y^2 = 1} \rightarrow 2y^2 + y^2 = 1 \rightarrow 3y^2 = 1 \rightarrow y^2 = \frac{1}{3}$$

~~$x=0 \rightarrow y=0$~~

$$\left(\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}}\right)$$

$$\left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}}\right)$$

$$\left(\sqrt{\frac{2}{3}}, -\sqrt{\frac{1}{3}}\right)$$

$$\left(-\sqrt{\frac{2}{3}}, -\sqrt{\frac{1}{3}}\right)$$

$$\boxed{y = \pm \sqrt{\frac{1}{3}}}$$

$$\underline{x} = \underline{1} 2y \rightarrow x^2 = 2y^2 = 2 \cdot \frac{1}{3} \rightarrow |x| = \pm \sqrt{\frac{2}{3}}$$

$$\underline{x^2 + y^2 = 1} \rightarrow 2y^2 + y^2 = 1 \rightarrow 3y^2 = 1 \rightarrow y^2 = \frac{1}{3}$$

$$\cancel{x=0} \rightarrow \cancel{y=0}$$

$$(\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}})$$

$$(-\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}})$$

$$\cancel{(\sqrt{\frac{2}{3}}, -\sqrt{\frac{1}{3}})} \leftarrow$$

$$\cancel{(-\sqrt{\frac{2}{3}}, -\sqrt{\frac{1}{3}})} \leftarrow$$

$$y = \pm \sqrt{\frac{1}{3}}$$

$$f(x, y) = \underline{x}^2 y$$

$$f(\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}}) = \frac{2}{3} \sqrt{\frac{1}{3}}$$

$$\underline{x}^2 = 2y \rightarrow x^2 = 2y^2 = 2 \cdot \frac{1}{3} \rightarrow |x| = \pm \sqrt{\frac{2}{3}}$$

$$\underline{x^2 + y^2 = 1} \rightarrow 2y^2 + y^2 = 1 \rightarrow 3y^2 = 1 \rightarrow y^2 = \frac{1}{3}$$

$$\cancel{x=0} \rightarrow \cancel{y=0}$$

$$(\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}})$$

$$(-\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}})$$

$$\cancel{(\sqrt{\frac{2}{3}}, -\sqrt{\frac{1}{3}})} \leftarrow$$

$$\cancel{(-\sqrt{\frac{2}{3}}, -\sqrt{\frac{1}{3}})} \leftarrow$$

$$y = \pm \sqrt{\frac{1}{3}}$$

$$f(x, y) = \underline{x}^2 y$$

$$f(\sqrt{\frac{2}{3}}, \sqrt{\frac{1}{3}}) = \frac{2}{3} \sqrt{\frac{1}{3}}$$